STATISTICS · UNIT 3: ANTICIPATING PATTERNS

Mystery Box Pop-Up

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design and analyze a $5 mystery box fundraiser. They build a probability model, simulate buyers, and use expected value to estimate profit.

CP1 and Honors add standard deviation, linear transformations, binomial and geometric models, combining random variables, sampling distributions, and the Central Limit Theorem.

Standards Addressed

- S-CP.A.1–5, S-CP.B.6–7 Probability rules, independence, conditional probability
- S-MD.A.1–4 Random variables, expected value, and probability distributions
- S-MD.B.5–7 Use expected values to evaluate decisions
- S-IC.A.2 Simulation; sampling distributions and the CLT

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: would you pay $5 for a mystery box? Part 1.
Day 2	Part 2 simulation or random variables.
Day 3	Part 3 and the business pitch to the “student council.”

Materials

- Student handout and Prize Cards
- Random digit table or calculator
- Optional: paper bags with colored slips to act out the boxes

How the Three Versions Differ

Level	What changes
CP2	Valid model, addition rule, simulation with digits, and expected value and profit.
CP1	Conditional probability and independence, mean and SD with transformations, combining two boxes, binomial and geometric.
Honors	Designing a prize table, differences of random variables, binomial probabilities, CLT for total profit, and sampling distribution of $\hat{p}$.

Watch For (Common Misconceptions)

- Adding standard deviations instead of variances.
- Thinking a price change changes the SD of profit.
- Using the binomial model when the number of trials is not fixed (geometric).
- Expecting a simulation of 10 buyers to match E(X).

Extension Ideas

- Simulate 1,000 days of sales with technology and compare with the CLT answer.
- Add a “second box half price” deal and find the new expected profit.
- Research loot box probabilities in video games.

Answer Key – CP2

Part 1: The Prize Model

#	Task	Answer
1	Show that the probabilities form a valid model.	0.50 + 0.30 + 0.15 + 0.05 = 1, and each is between 0 and 1.
2	Find P(headphones or gift card).	0.15 + 0.05 = 0.20
3	Find P(the prize is not a sticker).	1 − 0.50 = 0.50
4	Two boxes are bought. Find P(both are stickers).	0.50 × 0.50 = 0.25 (independent boxes)

Part 2: Simulate a Line of Buyers

#	Task	Answer
5	Use the digit ranges on the Prize Cards to find the prize for each of the 10 buyers.	Sticker, sticker, sticker, keychain, sticker, sticker, keychain, keychain, headphones, sticker
6	Find the total and the average prize value in your simulation.	Total $25; average $2.50
7	Did anyone win a gift card? Is that surprising for 10 buyers?	No; with P = 0.05, about 0.5 gift cards are expected in 10 boxes, so none is not surprising.

Part 3: Will We Make Money?

#	Task	Answer
8	Find E(X), the expected prize value per box.	1(0.50) + 3(0.30) + 10(0.15) + 25(0.05) = $4.15
9	What is the expected profit per $5 box?	5 − 4.15 = $0.85
10	How much profit should the council expect from 200 boxes?	200 × 0.85 = $170
11	Explain why the council could still lose money on a day it sells only a few boxes.	Expected value is a long-run average; with few boxes, a couple of gift cards could cost more than the sales.

Reflection

#	Prompt	Look for
R1	Why did your simulated average differ from $4.15?	Chance variation; the law of large numbers says the average gets close to $4.15 only over many boxes.
R2	Would you change the prizes? How would it change the expected profit?	Answers vary; e.g., raising the gift-card chance increases E(X) and lowers profit.

Answer Key – CP1

Part 1: Probability Rules

#	Task	Answer
1	Find P(headphones or gift card) and P(at least one gift card in 2 boxes).	0.20; $1 - 0.95^2 = 0.0975$
2	A buyer shakes the box and can tell the prize is worth more than \$2. Find P(gift card | worth more than \$2).	$0.05 \div 0.50 = 0.10$
3	Are "worth more than \$2" and "gift card" independent? Explain.	No; P(gift card | > \$2) = 0.10 ≠ P(gift card) = 0.05.
4	Describe how to use random digits to simulate how many boxes a buyer opens until getting headphones or a gift card.	Read pairs; 80–99 is a success; count pairs until the first success; repeat many times and average.

Part 2: Random Variables

#	Task	Answer
5	Find E(X) and SD(X).	E(X) = \$4.15; Var = 32.2275; SD ≈ \$5.68
6	Profit per box is Y = 5 − X. Find the mean and SD of Y.	Mean \$0.85; SD ≈ \$5.68 (unchanged)
7	The council raises the price to \$6. Find the new mean and SD of profit.	Mean \$1.85; SD ≈ \$5.68
8	Find the mean and SD of the total prize value T of two boxes.	Mean \$8.30; SD = $\sqrt{(2 \times 32.2275)} \approx \$8.03$

Part 3: Rare Prizes

#	Task	Answer
9	Explain why G is binomial and find its mean.	Fixed n = 20, two outcomes, independent, p = 0.05 each; mean = 20(0.05) = 1
10	Find P(G = 0) and P(G ≥ 1).	$0.95^{20} \approx 0.358$; $1 - 0.358 \approx 0.642$
11	A buyer keeps buying until getting a gift card. Find the probability the first gift card is in box 3, and the expected number of boxes.	$0.95^2(0.05) \approx 0.0451$; expected $1 \div 0.05 = 20$ boxes

Reflection

#	Prompt	Look for
R1	Why does changing the price change the mean profit but not the SD?	Adding a constant shifts every outcome equally, so the spread is unchanged.

#	Prompt	Look for
R2	Why do the variances (not SDs) add when combining two independent boxes?	For independent variables $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$; SDs do not add.

Answer Key – Honors

Part 1: Model the Box

#	Task	Answer
1	Find E(X), SD(X), and the mean and SD of Y.	E(X) = 4.15, SD ≈ 5.68; E(Y) = 0.85, SD(Y) ≈ 5.68
2	Find P(gift card | worth more than $2) and explain whether the events are independent.	0.10 ≠ 0.05, so not independent.
3	Two buyers each buy one box. Find the mean and SD of the difference of their prize values, $X_1 - X_2$.	Mean 0; SD = $\sqrt{(2 \times 32.2275)} \approx \$8.03$
4	Design a new prize table with E(X) = $4.00 that still has a $25 prize. Show it is valid.	Answers vary; one valid table: $1: 0.575, $3: 0.225, $10: 0.15, $25: 0.05. Sum = 1; E(X) = 0.575 + 0.675 + 1.5 + 1.25 = $4.00.

Part 2: Binomial and Geometric

#	Task	Answer
5	In 10 boxes, let H = number of headphones-or-gift-card prizes. Find P(H = 2) and P(H ≥ 3).	Binomial(10, 0.2): P(H = 2) ≈ 0.302; P(H ≥ 3) ≈ 0.322
6	Find the mean and SD of H.	Mean 2; SD = $\sqrt{(10 \times 0.2 \times 0.8)} \approx 1.26$
7	Find the probability a buyer gets a gift card within the first 10 boxes.	$1 - 0.95^{10} \approx 0.401$

Part 3: A Day at the Fair

#	Task	Answer
8	Find the mean and SD of total profit for 100 boxes.	Mean 100(0.85) = $85; SD = $\sqrt{100} \times 5.68 \approx \$56.77$
9	Use the CLT to find P(total profit < $0).	$z = (0 - 85) \div 56.77 \approx -1.50 \rightarrow \approx 0.067$
10	In 200 boxes, let $\hat{p}$ be the proportion with gift cards. Find its mean and SD and check the Large Counts condition.	Mean 0.05; SD = $\sqrt{(0.05 \times 0.95 \div 200)} \approx 0.0154$; np = 10 ≥ 10 and n(1 − p) = 190 ≥ 10 ✓
11	Find $P(\hat{p} \geq 0.08)$. Would 16 gift cards in 200 boxes make you suspect the box packer?	$z \approx 1.95 \rightarrow \approx 0.026$; yes, that would be unusual if p = 0.05.

Reflection

#	Prompt	Look for
R1	Why can we use a Normal model for total profit even though one box's profit is very skewed?	By the CLT, the sum (or mean) of many independent values is approximately Normal when n is large (n = 100).
R2	What is the trade-off between a bigger top prize and the risk of a losing day?	A bigger prize raises the SD (and maybe lowers the mean), which increases the chance of losing money.